

Objectives:

Task 1: Consider HCI or HRI?

- For each setting, consider: (a) who is the user, (b) what data does the model receive (manual entry, EHR pull, device stream), (c) what does the model emit (a number, a colour-coded alert, an audio cue, a robot behaviour), (d) what does the human do next.
- Complete co-design canvas on “Problem Space” for both contexts (from canvas template in GitHub - see resources).
- Compare the HCI and HRI implementation: ~3 HCI-specific safety concerns (e.g. alarm fatigue, accessibility, display legibility under cover of stress) vs ~3 HRI-specific safety concerns (e.g. proximity safety, voice input under noisy ED conditions, graceful degradation if sensors fail).

Status: Completed.

Task 2: Co-Design Canvas

- Complete the canvas for your preferred implementation.
- Ethics, Guidelines, MVP, Environment, Form (from canvas template in GitHub - see resources).

Status: Will be completed for Final Submission (Aug 4).

Task 3: Mock-Ups

- Produce rough mock-ups using any tool(s) of your choice (web applications - wire frame via Canva/Figma and physical robot - TinkerCAD/Blender).
- Note that robotic applications may use both HCI and HRI principles.

Status: Will be completed for Final Submission (Aug 4).

Task 4: System Requirements Document

- Outline requirements for the system in a document (these should be reflected in task 1 and 2).

Status: Will be completed for Final Submission (Aug 4).

Task 5: Walk-Through Video

- Record a 2-minute video walking through the vision for your system.
- Post on to Discord.

Status: Will be completed for Final Submission (Aug 4).

Interim Submission (Saturday 11:59 p.m. AST)

- Problem space co-design canvas for HRI and HCI.
- Co-design canvas for your desired method (at least Problem, Ethics, Environment sections filled).
- Initial mock-up sketches.
- Notes on system requirements (inputs, outputs, and integration requirements).

Link to GitHub Repo: <https://github.com/sarianaramoutar/carisurg-portfolio.git>

Task 1: Consider HCI or HRI?

Dimension	Setting A Main ED Triage Desk (HCI)	Setting B Observational Unit Intake Robot (HRI)
Target User	ED triage nurses (e.g., Sister Patrice Alleyne's team) working in a high-pressure environment where rapid, safe decisions are required.	Patients arriving in the Observation Unit, supported by observation nurses who supervise the interaction and receive the collected information.
Input Data and Route	Primary: Automatic Electronic Health Record (EHR) retrieval combined with manually entered presenting complaint, pain score and any missing clinical information. Fallback: Fully manual entry if EHR connectivity is unavailable.	Primary: Robot-guided patient questionnaire, integrated vital-sign sensors (blood pressure, heart rate, oxygen saturation, temperature, glucose where available), and touchscreen selection of existing medical conditions. Fallback: Observation nurse manually completes the assessment if sensors fail or the patient cannot use the kiosk independently.
Model Output	Predicted ESI level (1–5), confidence score, colour- and icon-coded urgency indicator, and the top clinical factors contributing to the prediction to support explainability.	Confirmation that patient assessment has been completed, safe patient guidance through the interface, and secure transmission of all collected data to the nurse dashboard. If abnormal observations are detected, an immediate alert is sent to clinical staff.
Human Action Required	Nurse reviews the AI recommendation alongside the explanation, accepts or overrides the suggested ESI level, records the reason for any override (to assist with future training and for documentation purposes), and proceeds with appropriate triage and treatment.	Patient follows on-screen and/or voice-guided instructions while the robot collects information. Observation nurses monitor incoming assessments, intervene whenever required, and continue clinical assessment using the transmitted information.

The HCI system improves **decision making** while the HRI system improves **data collection**.

HCI Problem Space

Clinical Problem

Emergency Department triage nurses at Mercer General often assess more than 40 patients during a 12-hour shift. During periods of high patient volume, triage decisions must frequently be made within three minutes using incomplete clinical information. Distinguishing patients with moderate acuity (particularly ESI Levels 3–5) can be challenging, increasing the risk of inconsistent triage decisions, delayed intervention, and inefficient allocation of clinical resources.

Who experiences the problem?

- ED triage nurses working in a fast-paced, high-pressure environment.
- Patients whose condition may not initially appear critical, particularly those within ESI Levels 3–5.
- The wider Emergency Department through increased waiting times and workflow disruption.

Current Manual Workaround

Triage nurses combine clinical judgement, patient history, vital signs and established triage protocols while manually documenting assessments. During busy periods, maintaining consistency across similar patient presentations becomes increasingly difficult and contributes to cognitive workload and decision fatigue.

System Solution Value

The Explainable AI Assistant, powered by an optimised Random Forest model, provides a real-time ESI classification together with a confidence score and the three most influential clinical factors contributing to the prediction. The system highlights potential inconsistencies and supports evidence-based decision making while ensuring the nurse retains full authority to accept or override every recommendation.

HCI Safety Considerations

1. Alarm Fatigue & Visual Overload

Concern

Frequent alerts or unnecessary notifications may cause triage nurses to ignore or dismiss AI recommendations, reducing the effectiveness of the decision-support system.

Context

During busy 12-hour shifts, nurses manage multiple patients while responding to monitor alarms, medication reminders and other clinical notifications. Excessive AI alerts could become background noise rather than useful decision support.

Mitigation

Only display high-priority alerts when the Explainable AI model identifies a clinically significant disagreement with the nurse's initial triage decision (for example, the nurse selects ESI 4 while the model strongly predicts ESI 2). Lower-confidence recommendations should appear as passive visual prompts rather than interruptive alerts.

Residual Risk

During major incidents or periods of extreme workload, even important alerts may be overlooked or intentionally bypassed.

2. Display Legibility & Accessibility**Concern**

Critical patient information may be difficult to interpret quickly because of poor lighting, screen glare, visual fatigue or colour vision deficiency.

Context

The triage dashboard must remain readable under fluorescent lighting, during overnight shifts and in high-pressure situations where nurses only have seconds to scan the display.

Mitigation

Use large, high-contrast text, WCAG-compliant colour combinations, and ensure every colour-coded alert is reinforced with icons, text labels and clear visual hierarchy. Important information, including the predicted ESI level and top contributing factors, should always appear in the same location on the screen.

Residual Risk

Individual differences in eyesight, stress levels or damaged display hardware may still reduce readability despite accessibility-focused design.

3. Power, Connectivity & System Resilience**Concern**

Power interruptions, network outages or temporary Electronic Health Record (EHR) failures may interrupt the AI-assisted triage workflow or prevent patient data from synchronising.

Context

Caribbean healthcare facilities may experience generator changeovers, temporary connectivity loss or network congestion during periods of high demand.

Mitigation

The system should automatically preserve incomplete assessments locally, continue operating in offline mode where possible, and allow nurses to complete manual data entry until EHR connectivity is restored. Any unsynchronised records should upload automatically once the connection becomes available.

Residual Risk

During prolonged outages, real-time patient records may temporarily become inconsistent across different hospital systems, requiring manual verification by staff.

HRI Problem Space

Clinical Problem

Patients arriving in the Observation Unit may experience delays before an initial assessment can begin, particularly when nursing staff are responding to emergencies elsewhere in the department. During this waiting period, valuable clinical information including presenting complaints, medical history and baseline vital signs may not be collected immediately, delaying the identification of patients whose condition may be deteriorating.

Who experiences the problem?

- Patients waiting for assessment in the Observation Unit.
- Observation nurses responsible for monitoring multiple patients simultaneously.
- Clinical staff coordinating patient flow throughout the Emergency Department.

Current Workaround

Patients wait until an observation nurse becomes available to manually collect demographic information, presenting complaints and vital signs using portable monitoring equipment. During periods of high workload, repeated reassessment may be delayed as nurses prioritise more urgent clinical responsibilities.

System Solution Value

An accessible robotic intake assistant guides patients through a structured assessment by collecting demographic information, chief complaints, relevant medical history and baseline vital signs. The information is securely transmitted to the Explainable AI platform, allowing the model to generate an evidence-based ESI prediction for review by nursing staff. Any abnormal findings are immediately highlighted to clinicians, while all final triage decisions remain the responsibility of the nurse.

HRI Safety Considerations

1. Physical Proximity & Patient Safety

Concern

The robot operates close to patients who may have reduced mobility, cognitive impairment or acute illness, creating a risk of accidental contact, anxiety or injury.

Context

The Observation Unit is a shared clinical environment with wheelchairs, mobile equipment, nurses and patients moving through confined spaces.

Mitigation

The robot should remain largely stationary during assessments, use obstacle and proximity sensors, limit any physical movement, and include an easily accessible emergency stop button. Staff should be able to immediately disable the robot if required.

Residual Risk

Some patients may still become uncomfortable or startled simply because of the robot's physical presence, particularly if they are unfamiliar with healthcare robotics.

2. Voice Input & Accessible Patient Interaction

Concern

High ambient noise or patient communication difficulties may result in incorrect symptom recording or unsuccessful voice interactions.

Context

Emergency Departments are noisy environments with alarms, conversations, equipment and generator noise. In addition, some patients may have hearing impairments, speech impairments or strong regional accents.

Mitigation

Voice interaction should be optional rather than mandatory. The primary interface should be a large touchscreen with multilingual support, adjustable text size, simple navigation and confirmation screens before information is submitted. Voice responses should always require visual confirmation.

Residual Risk

Some patients may still require assistance from a nurse because of severe illness, limited digital literacy or communication difficulties.

3. Sensor Failure & Clinical Data Reliability

Concern

Vital-sign sensors may produce inaccurate or incomplete measurements because of patient movement, poor sensor contact, hardware malfunction or environmental conditions.

Context

The robot collects physiological data before a clinician has independently assessed the patient. Incorrect measurements could influence the AI model if they are not identified.

Mitigation

Each vital sign should undergo automatic quality checks before being accepted. If measurements fall outside expected ranges or sensor confidence is low, the system should clearly notify staff, prevent an AI prediction from being generated using incomplete data, and request manual verification before continuing.

Residual Risk

Minor inaccuracies that remain within clinically plausible ranges may not always be detected automatically and may still require professional judgement during the nurse's assessment.

Co-design Canvas

PROBLEM SPACE (HCI) - ISSUE		
GROUP(S)	CHARACTERISTICS	NEEDS
Primary { • ED triage nurses • Nurses supervising patient flow	• fast-paced, high pressure enviro. • quick decisions (~ 3 mins)	• fast consistent triage support • explainable AI recommendations
Secondary { * Emergency physicians * Admin staff * Patients	• frequent interruptions + cognitive overload • require reliable, easy to interpret info * depend on timely & accurate triage decisions * have varying levels of technical knowledge * need info relevant to their specific role	• ability to override AI decisions • minimal disruption to existing workflow * accurate patient prioritisation * efficient patient flow * reliable clinical documentation * safe communication of patient info
GOALS		
• complete accurate triage quickly • identify patients requiring urgent attention • make confident, evidence-supported decisions * receive timely & accurate patient info * improve coordination between clinicians	• reduce decision fatigue • improve consistency across shifts • build trust in explainable AI decision support * improve ED efficiency * reduce waiting times * improve patient safety & quality of care	

ETHICAL CONSIDERATIONS (HCI) etc

Physical Safety

- | | | |
|-----------|--|--|
| user | • Incorrect AI recommendations could contribute to inappropriate usage if accepted without reviewing | • The AI functions only as decision support. Final clinical responsibility remains with the nurse. |
| AI system | * Incorrect/incomplete patient data may produce unreliable predictions | * System flags incomplete data & requests manual verification before generating recommendations |
| | <u>PROBLEM</u> | <u>SOLUTION</u> |

Data Security

- | | | |
|-----------|--|--|
| user | • Patient info could be viewed by unauthorised individuals at the time of use. | • Encrypted data transmission, role-based access and privacy screen filters |
| AI system | * Encrypted data transmission, role-based access
Secure storage of patient assessments and override logs | * Encrypted storage with audit trails and compliance with hospital governance policies |
| | <u>PROBLEM</u> | <u>SOLUTION</u> |

Transparency

- | | |
|---|--|
| • Users may not understand how the AI reached its recommendations | • Display the predicted ESI level, confidence score & top contributing factors |
| * The AI may be perceived as replacing clinical judgement | * Clearly label outputs as AI Recommendations and emphasise nurses make the final decision |

Equality across users

- | | |
|---|--|
| • Model performance may vary across diff patient populations because original training data were not Caribbean-specific | • Monitor performance across demographics & retrain model using local hospital data when available |
| * Algorithmic bias may reduce fairness | * Regular bias audits & continuous monitoring of override patterns |

<u>Foreseen Concerns</u>	
- Some nurses may distrust / over-rely on AI recommendations * The AI should not encourage overconfidence in its predictions	- Use explainable outputs & maintain clinician control through the override function * Display confidence levels & encourage additional review for low-confidence predictions
<u>Behaviour Enforcement</u>	
- Nurses may ignore AI recommendations / rely on them without sufficient assessment. * The AI should encourage responsible clinical use.	- Provide training on appropriate use & reinforce clinical support - not replacement. * Record overrides with documented reasons to support quality improvement & future model refinement.

<u>ENVIRONMENT (HCI)</u>	
<u>WHERE</u>	<u>USERS</u>
Main ED image desk at Mercer General	ED image nurses
<u>WHEN</u>	<u>SECONDARY USERS</u>
24 hr / day across day & night shifts, with high workload during peak periods.	Emergency physicians, administrative staff, nurse supervisors.
<u>DATA COLLECTION</u>	<u>SIMULTANEOUS USERS</u>
System receives patient info demographics, chief complaint, vital signs, relevant clinical info from the EHR & manual nurse entry. All data stored in hospital info system.	(Few) Multiple nurses may use separate workstations simultaneously, but each patient assessment is completed by one nurse.
<u>EXTERNAL SENSORS & ACTUATORS</u>	<u>CONNECTION TO SYSTEMS</u>
Receives data from connected vital signs monitors (hr, bp, SpO ₂ , temp). No physical actuation required for HCI implementation.	Integrated with hospital Electronic Health Record (EHR), image software & vital monitoring devices. Predictions & override logs are securely stored to support documentation & future model improvement.

NB. I apologise for the handwriting. Will do a cleaner version for the Final Submission.

MERCER GENERAL
EMERGENCY DEPARTMENT

EXPLAINABLE AI TRIAGE ASSISTANT

AI recommendations support, but do not replace, clinical judgement.

Nurse: P. Alleyne
Triage Nurse

01 Aug 2026
03:17 a.m.

PATIENT INFORMATION

Patient ID
MG-2026-0891

Age
64

Sex
Male

Chief Complaint
Chest Pain

VITAL SIGNS

Heart Rate
112 bpm

Blood Pressure
165 / 95 mmHg

Oxygen Saturation (SpO₂)
91 %

Temperature
37.8 °C

Pain Score
8 / 10

EXPLAINABLE AI RECOMMENDATION

PREDICTED ESI LEVEL

2

ESI LEVEL 2
EMERGENT

CONFIDENCE

89%

HIGH

WHY WAS THIS RECOMMENDED? ⓘ

1 Oxygen saturation below 92%
High impact

2 Heart rate above 110 bpm
High impact

3 Age above 60 years
Moderate impact

CLINICAL RECOMMENDATION

Review immediately.
Consider oxygen therapy and urgent physician assessment.

ACCEPT RECOMMENDATION
Assign ESI Level 2

OVERRIDE
Choose a different ESI level

IMPORTANT: This recommendation supports, but does not replace, clinical judgement.
Final triage decision remains the responsibility of the nurse.

TRIAGE QUEUE SUMMARY

ESI LEVEL	COUNT	STATUS
1	0	Critical
2	2	High Priority
3	8	Moderate
4	14	Low Priority
5	5	Lowest Priority

LIVE TRIAGE QUEUE

PATIENT ID	ESI	STATUS
# 101	2	Waiting
# 102	4	Waiting
# 103	3	Being Assessed
# 104	5	Waiting
# 105	2	Waiting

[View full queue →](#)

System Status
All systems operational

Connection
EHR Connected

Data Security
Encrypted & Compliant

Override Logging
All overrides are recorded for documentation and model improvement